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DEEP LEARNING MINI PROJECT REPORT(CS722T2C) on

“Telco Customer Churn Prediction”

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In partial fulfillment of the requirements for the VII semester

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE & ENGINEERING

Under the Guidance of

Ms. Shiji Abraham

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at



SAHYADRI

College of Engineering & Management

An Autonomous Institution

MANGALURU

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EVALUATION RUBRICS

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Sl. No.	Evaluation Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)	Marks
1	Problem Definition & Objectives	Problem is clearly defined, relevant, and objectives are specific and measurable	Problem and objectives are clear with minor gaps	Problem is partially defined; objectives need improvement	Problem/objectives are unclear	
2	Technical Approach & Implementation	Appropriate methodology, algorithms/tools used effectively; implementation is complete	Appropriate technical approach with minor issues	Basic implementation with limited technical depth	Incomplete or inappropriate implementation	
3	Innovation / Application / Problem Solving	Demonstrates strong originality, creativity, and effective solution to the identified problem	Shows some innovation and good problem-solving	Limited innovation; mostly conventional approach	No significant innovation/problem-solving	
4	Results, Testing & Analysis	Results are clearly presented, thoroughly tested, and critically analyzed using suitable measures	Results presented with adequate testing and analysis	Basic results and limited testing/analysis	Results are incomplete or unsupported	
5	Documentation, Presentation & Viva	Well-structured report, clear presentation, excellent communication and confident viva	Good documentation, presentation and satisfactory viva	Adequate documentation/presentation with limited clarity	Poor documentation, presentation or viva	
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Department of Computer Science & Engineering

DECLARATION

We hereby declare that the entire work embodied in this Mini Project Report titled “**Telco Customer Churn Prediction**” has been carried out by us at Sahyadri College of Engineering & Management, Mangaluru under the supervision of **Ms. Shiji Abraham**, in partial fulfillment of the requirements for the VII semester of **Bachelor of Engineering in Computer Science & Engineering**. This report has not been submitted to this or any other University for the award of any other degree.

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Abstract

Customer churn is a major challenge for telecom companies because losing customers reduces recurring revenue and acquiring new ones is more expensive than retaining existing ones. This project develops a **Telco Customer Churn Prediction System** using deep learning to identify customers who are likely to leave, enabling proactive retention strategies. The system uses the **IBM Telco Customer Churn dataset** containing 7,043 customer records and 19 input attributes. Data is cleaned and transformed through a scikit-learn preprocessing pipeline into 30 model features. A deep neural network with three dense layers, batch normalization, and dropout predicts churn probability, while a decision threshold of **0.45** classifies customers into churn or non-churn categories. Customers are further grouped into **Low, Medium, and High** risk levels to support retention planning. **SHAP** explanations provide transparency by identifying the key factors influencing each prediction. The model is deployed through a **Flask API** and integrated with a **Streamlit dashboard** for single-customer prediction, batch analysis, explanation generation, and downloadable reports. On the test dataset, the model achieved **79.70% accuracy**, **62.29% precision**, **59.63% recall**, and **84.19% ROC-AUC**. Key churn drivers include tenure, monthly charges, fiber-optic internet, contract type, and payment method, making the system an effective and explainable decision-support tool.

Keywords: customer churn, deep neural network, explainable AI, SHAP, telecommunications analytics, Streamlit, Flask

Table of Contents

Content	Page No.
Abstract	i
Table of Contents	ii
1. Introduction	1
1.1 Overview	1
1.2 Scope & Motivation	1
2. Literature Survey	3
3. Problem Formulation	8
3.1 Problem Description	8
3.2 Problem Statement	8
3.3 Objectives	8
3.4 Functional Requirements	9
3.5 Non-Functional Requirements	10
4. Project Design and Implementation	12
4.1 Proposed Project Architecture	12
4.2 High Level Design	15
4.3 Detailed Design	18
4.4 Datasets Descriptions	23
4.5 Tools and Technologies used for Implementation	25
5. Results and Discussion	27
5.1 Experimentation Environment Set up	27
5.2 Functional Requirements Results achieved	28
5.3 Non-Functional Requirements Results achieved	35
5.4 Comparison of Results with Existing works	36
5.5 Project GUI Snapshots (Major Functionalities)	38
5.6 Societal Impact of the Project	40
5.7 SDG Mapped	41
Conclusion and Future Scope	42
References	44

CHAPTER 1

Introduction

Customer churn is a major challenge for telecommunication companies as it leads to revenue loss and reduced customer retention. This project develops a Deep Learning-based Customer Churn Prediction System to identify customers who are likely to leave using historical customer data. It also integrates SHAP-based Explainable AI to provide transparent predictions and support proactive customer retention strategies.

1.1 Overview

Customer retention has become a critical concern for telecommunication companies due to intense market competition and the high cost of acquiring new customers. Losing existing customers, commonly referred to as customer churn, results in reduced revenue and negatively impacts long-term business growth. Traditional methods of identifying customers who are likely to churn are often manual, time-consuming, and ineffective for handling large volumes of customer data. Therefore, there is a growing need for intelligent systems that can accurately predict churn and enable organizations to take proactive retention measures.

This project presents a **Deep Learning-based Telco Customer Churn Prediction System** that analyzes customer information and predicts the likelihood of churn before it occurs. The proposed system performs data preprocessing, feature engineering, and churn prediction using a Deep Neural Network (DNN). To improve transparency and trust in the model, **SHAP (SHapley Additive exPlanations)** is integrated to explain the key factors influencing each prediction. The application is developed using **Streamlit** for the user interface and **Flask** for backend API integration, supporting both single-customer and batch predictions with downloadable reports. This intelligent framework assists telecom companies in identifying high-risk customers, understanding the reasons behind churn, and implementing timely retention strategies.

1.2 Scope & Motivation

The scope of this project is to develop a user-friendly and intelligent system that predicts customer churn using deep learning and provides clear explanations for every prediction. The system analyzes customer demographic and service-related data by performing preprocessing techniques such as handling missing values, encoding categorical features,

and standardizing numerical data. Based on the prediction, customers are classified into **Low**, **Medium**, or **High** churn risk categories, with SHAP explanations highlighting the factors that influence each result. The application also supports both single-customer and batch predictions, making it suitable for real-world telecom business operations.

The motivation for this project comes from the growing challenge of customer churn in the telecommunications industry. Losing customers not only reduces revenue but also increases the cost of acquiring new ones. Instead of identifying churn after it happens, this system helps companies detect customers who are likely to leave in advance. By combining deep learning with explainable AI, businesses can better understand customer behavior, take timely retention actions, improve customer satisfaction, and make more informed, data-driven decisions.

CHAPTER 2

Literature Survey

Before designing the system we reviewed published work on telecom churn prediction to see what has been tried, what worked and where the gaps are. The studies fall into a few loose groups. A large body of work uses classical and ensemble machine-learning models such as decision trees, random forests, gradient boosting and stacking [1], [7], [12]–[14], [18]. A second group applies neural and deep-learning models to behavioural or tabular customer data [2], [4], [16], [21]. More recent studies add explainability, most often SHAP, so that predictions can be justified to business users [10], [19], [22], [23]. A few review papers pull the field together and point out open problems [15], [20].

Table 2.1 summarises the method, contribution and limitation of each of the reviewed studies.

Table 2.1: Literature Review Summary

Ref	Year	Author(s)	Methodology	Key Contribution	Limitations
[1]	2019	Abdelrahim Kasem Ahmad, Assef Jafar, Kadan Aljoumaa	Used ML algorithms (Decision Tree, Random Forest, GBM, XGBoost) with big data processing and feature selection for telecom churn prediction.	Developed a scalable churn prediction model using telecom big data and improved prediction performance.	Requires large-scale infrastructure and lacks explainable AI methods.
[2]	2020	Nadia Alboukaey et al.	Applied deep learning techniques on customer behavioral patterns and sequential telecom data for churn prediction.	Captured complex customer behavior patterns for better churn identification.	Requires large datasets and high computational resources.
[3]	2020	Huang et al.	Applied data mining and classification techniques for telecom customer churn prediction.	Identified important factors influencing customer churn behavior.	Older datasets may not represent current telecom trends.

Ref	Year	Author(s)	Methodology	Key Contribution	Limitations
[4]	2021	H. Nalatissifa and H. F. Pardede	Built a deep neural network for Telco churn data and used an automated search over the number of hidden layers, dropout rate and learning rate, comparing activation functions and optimizers.	Showed that a carefully tuned DNN can be competitive with traditional models on churn data.	Not automatically better than classical models; results depend on careful tuning.
[5]	2021	Idris et al.	Used cost-sensitive learning methods to handle class imbalance in churn datasets.	Improved detection of minority churn customers	Requires additional model tuning and complexity.
[6]	2021	Ahmed A. Alhaj , Mohamed M. A. Salama , Abdulla Alarood, Sayed Abdel-Khalek	Used feature selection methods with SVM, Random Forest, MLP, and Naive Bayes classifiers.	Improved model accuracy by selecting important customer attributes.	Additional computational processing is required for feature selection.
[7]	2021	Tian Xu, Y. Ma, K. Kim	Proposed an ensemble learning approach with feature grouping techniques for telecom churn classification.	Improved prediction accuracy by combining multiple machine learning models.	Ensemble models increase complexity and reduce interpretability.
[8]	2022	De Caigny et al.	Applied interpretable machine learning methods for customer churn prediction.	Improved understanding of factors affecting customer churn decisions.	Interpretability may reduce prediction performance.
[9]	2022	Authors of MDPI Information	Applied machine learning classification algorithms using customer demographic and service usage data.	Demonstrated effectiveness of ML models for predicting telecom customer churn.	Limited exploration of deep learning and explainability methods.

Ref	Year	Author(s)	Methodology	Key Contribution	Limitations
[10]	2022	Peng K., Peng Y.	Developed GA-XGBoost based churn prediction model integrated with SHAP explainability techniques.	Combined optimization and explainable AI to identify major churn factors.	Evaluated on limited datasets and requires further validation.
[11]	2023	Chen et al.	Proposed hybrid machine learning models combining multiple classifiers.	Increased prediction reliability using hybrid learning techniques.	Combined models are difficult to interpret.
[12]	2023	Victor Chang et al.	Applied ensemble machine learning models for telecom customer churn prediction.	Improved churn classification using combined learning approaches.	Performance depends on dataset quality and feature selection.
[13]	2024	Elsevier Researcher s	Developed stacking ensemble learning models using multiple ML classifiers for telecom churn prediction.	Improved prediction accuracy through hybrid ensemble approaches	Requires extensive parameter optimization
[14]	2024	IEEE Researcher s	Compared Gradient Boosting, CatBoost, AdaBoost, and Random Forest algorithms for churn prediction.	Demonstrated effectiveness of ensemble learning approaches.	Limited focus on explaining prediction decisions.
[15]	2024	Springer SN Computer Science Researcher s	Reviewed machine learning and deep learning techniques used for telecom churn prediction.	Identified research trends, datasets, algorithms, and future research directions.	Does not provide experimental implementation of a new model.

Ref	Year	Author(s)	Methodology	Key Contribution	Limitations
[16]	2024	Somak Saha, Chamak Saha, Md. Mahidul Haque, Md. Golam Rabiul Alam, Ashis Talukder	Developed ChurnNet using Deep Neural Network with preprocessing and feature engineering.	Achieved improved prediction performance compared with traditional ML models.	Requires high computational resources and lacks explainability.
[17]	2024	Authors of MDPI Algorithms	Proposed an optimized AI framework using machine learning algorithms for churn prediction.	Improved prediction accuracy through optimization techniques.	Model interpretation for business decisions was not deeply explored.
[18]	2024	Sharmila K. Wagh et al.	Compared Random Forest, Decision Tree, and KNN classification algorithms for churn prediction.	Identified effective ML algorithms for customer retention analysis.	Limited focus on advanced AI models and real-time deployment.
[19]	2024	Sumana Sharma Poudel et al.	Used machine learning models with SHAP-based explainable AI techniques for telecom churn prediction.	Improved transparency by explaining reasons behind churn predictions.	Mainly focused on structured datasets.
[20]	2025	Springer Researchers	Surveyed AI, ML, DL, and XAI techniques for telecom churn prediction.	Provided future research directions including explainable AI.	Review-based study without experimental implementation.
[21]	2019	Y. Khan, S. Shafiq, A. Naeem, S. Hussain, S. Ahmed, N. Safwan	Applied artificial neural networks to telecom customer attributes covering demographic, billing and usage information.	Reached about 79% accuracy and helped surface the factors most associated with customer attrition.	Moderate accuracy; findings are specific to the dataset used.

Ref	Year	Author(s)	Methodology	Key Contribution	Limitations
[22]	2025	H. A. Rasool, K. K. Aljawaheri, A. A. M. Karram	Stacked Decision Tree, Random Forest, Extra Trees, Gradient Boosting and XGBoost with SMOTE, and explained results with SHAP.	Reported 94.75% accuracy; SHAP highlighted total day minutes, international plan and account length.	Evaluated on a different telecom dataset, so results are not directly comparable; stacking adds complexity.
[23]	2026	A. El Attar and M. El-Hajj	Trained seven ML models (including XGBoost and an MLP) on the Telco dataset with feature engineering and SMOTE; used SHAP and customer segmentation.	XGBoost gave the best ranking ability (AUC 0.932); SHAP pointed to contract type, tenure and technical support.	Focused on offline modelling and analysis rather than an interactive tool.

Conclusion

This literature review shows that machine learning, deep learning, and ensemble learning techniques have significantly improved the accuracy of telecom customer churn prediction. Recent studies have also introduced Explainable Artificial Intelligence (XAI) methods, such as SHAP, to make prediction results more transparent and useful for business decision-making. These approaches help telecom companies identify customers who are likely to leave and implement effective retention strategies.

However, existing research still faces challenges, including limited model interpretability, high computational requirements, class imbalance, and dependence on high-quality datasets. Based on these research gaps, the proposed project aims to develop an accurate, explainable, and user-friendly customer churn prediction system by integrating machine learning and SHAP explainability to provide meaningful insights and personalized customer retention recommendations.

CHAPTER 3

Problem Formulation

This chapter defines the core problem addressed by this project, outlines the specific problem statement, and establishes the objectives that guide the system design and implementation.

3.1 Problem Description

The telecommunications industry is highly competitive, making customer retention a major challenge. Many customers discontinue telecom services due to factors such as high service costs, poor network quality, inadequate customer support, and attractive offers from competitors. Customer churn directly affects the revenue and profitability of telecom companies, as acquiring new customers is more expensive than retaining existing ones.

Traditional churn prediction methods often fail to identify customers at risk in advance and provide little explanation for their predictions. Moreover, business managers require transparent and interpretable insights to understand the reasons behind customer churn and take appropriate retention actions. Therefore, there is a need for an intelligent system that can accurately predict customer churn, explain the prediction results, and assist decision-makers in designing effective customer retention strategies.

3.2 Problem Statement

Existing telecom customer churn prediction systems primarily focus on improving prediction accuracy but often lack explainability and user-friendly decision support. The absence of interpretable predictions makes it difficult for telecom companies to understand the factors influencing customer churn and implement personalized retention strategies. Therefore, there is a need to develop an intelligent customer churn prediction system that integrates machine learning, explainable AI (SHAP), and an AI-powered chatbot to accurately predict customer churn, explain prediction outcomes, and provide actionable retention recommendations.

3.3 Objectives

The objectives of this project are as follows:

1. To design and develop a machine learning-based telecom customer churn prediction system using customer demographic, subscription, and service usage data.
2. To implement an explainable machine learning model integrated with SHAP (SHapley Additive exPlanations) for accurate churn prediction and interpretation of prediction results.
3. To develop a user-friendly interface with an AI chatbot for churn prediction, explainable insights, and personalized customer retention recommendations.

3.4 Functional Requirements

FR-01: Customer Data Input

- The system shall accept telecom customer information, including demographic, account, and service usage details.
- The system shall validate customer data before using it for churn prediction analysis.

FR-02: Data Preprocessing

- The system shall perform data cleaning, missing value handling, encoding, feature scaling, and feature selection before prediction.
- The system shall transform raw customer data into a suitable format for machine learning model processing.

FR-03: Customer Churn Prediction

- The system shall classify customers as **Likely to Churn** or **Not Likely to Churn** using a trained machine learning model.
- The system shall provide churn risk prediction results based on customer service and usage patterns.

FR-04: Explainable AI Analysis

- The system shall generate SHAP explanations to identify the most influential features contributing to churn prediction.
- The system shall display feature importance information to improve understanding of prediction outcomes.

FR-05: AI Chatbot Assistance

- The system shall provide an AI chatbot that explains prediction results in simple language and suggests personalized customer retention strategies.
- The chatbot shall assist users by providing recommendations to reduce customer churn.

FR-06: Performance Evaluation

- The system shall evaluate the prediction model using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix.
- The system shall generate evaluation results to measure model performance and reliability.

3.5 Non-Functional Requirements

NFR-01: Performance

- The system shall load the model, preprocessor and explainer once per running process instead of on every request.
- The system shall return a single-customer prediction interactively and score an uploaded CSV file in one pass.

NFR-02: Reliability and Reproducibility

- The system shall apply the same saved preprocessing pipeline that was used during training, so identical inputs always give identical outputs.
- Reported metrics shall be tied to a documented evaluation protocol (split, seed and threshold).

NFR-03: Usability

- The interface shall be usable by retention staff without data-science training, using simple forms, plain-language explanations, ranked tables and downloadable reports.

NFR-04: Data Integrity and Input Protection

- The system shall check required fields, data types, allowed category values and CSV structure before any record reaches the model.
- The customer identifier shall be kept for record keeping but excluded from the model inputs.

NFR-05: Maintainability and Configurability

- Presentation, validation, inference and explanation shall be kept in separate layers so that one can change without touching the others.
- The classification threshold and risk bands shall be configurable without retraining the model.

NFR-06: Portability

- The system shall run on an ordinary CPU-based machine, without a GPU, on Python 3.9 to 3.13.

NFR-07: Transparency

- Every prediction shall be accompanied by SHAP explanations, at both individual-customer and portfolio level.
- Explanations shall be presented as descriptions of model behaviour, not as proof of cause.

CHAPTER 4

Project Design and Implementation

This chapter explains how the system is built. It starts with the overall architecture, moves to the modules and data flow, and then goes into the detailed design of the preprocessing pipeline, the neural network, the decision policy and the explanation module. It ends with a description of the dataset and the tools used.

4.1 Proposed Project Architecture

The system is organised as a stack of layers, and each layer has a single responsibility. Figure 4.1 follows one request from the user to the final output, and Figure 4.2 shows the same design as a layer diagram.

A user enters a customer's details in the Streamlit dashboard, either through a form or by uploading a CSV file. The dashboard does no modelling of its own; it forwards the data to the Flask REST API. The API first validates the record against the expected schema of 19 input fields. Valid records are then processed using the saved scikit-learn preprocessor, which transforms the input into the 30 features required by the machine learning model. The processed data is passed to the saved TensorFlow neural network, which predicts the probability of customer churn. The predicted probability is then used for two parallel tasks: assigning a churn category and risk level based on predefined thresholds, and generating a SHAP-based local explanation to identify the most influential features affecting the prediction. The results from both processes are combined and returned to the Streamlit dashboard, where they are displayed as interactive visualizations, downloadable HTML reports, or ranked CSV files for batch predictions. The dashboard also presents feature importance graphs to help users understand the key factors contributing to customer churn. In addition, an integrated AI chatbot explains the prediction results in simple language and provides personalized customer retention recommendations based on the customer's profile. This layered architecture improves modularity, scalability, maintainability, and system reliability by allowing each layer to perform a specific function while communicating efficiently with other components.

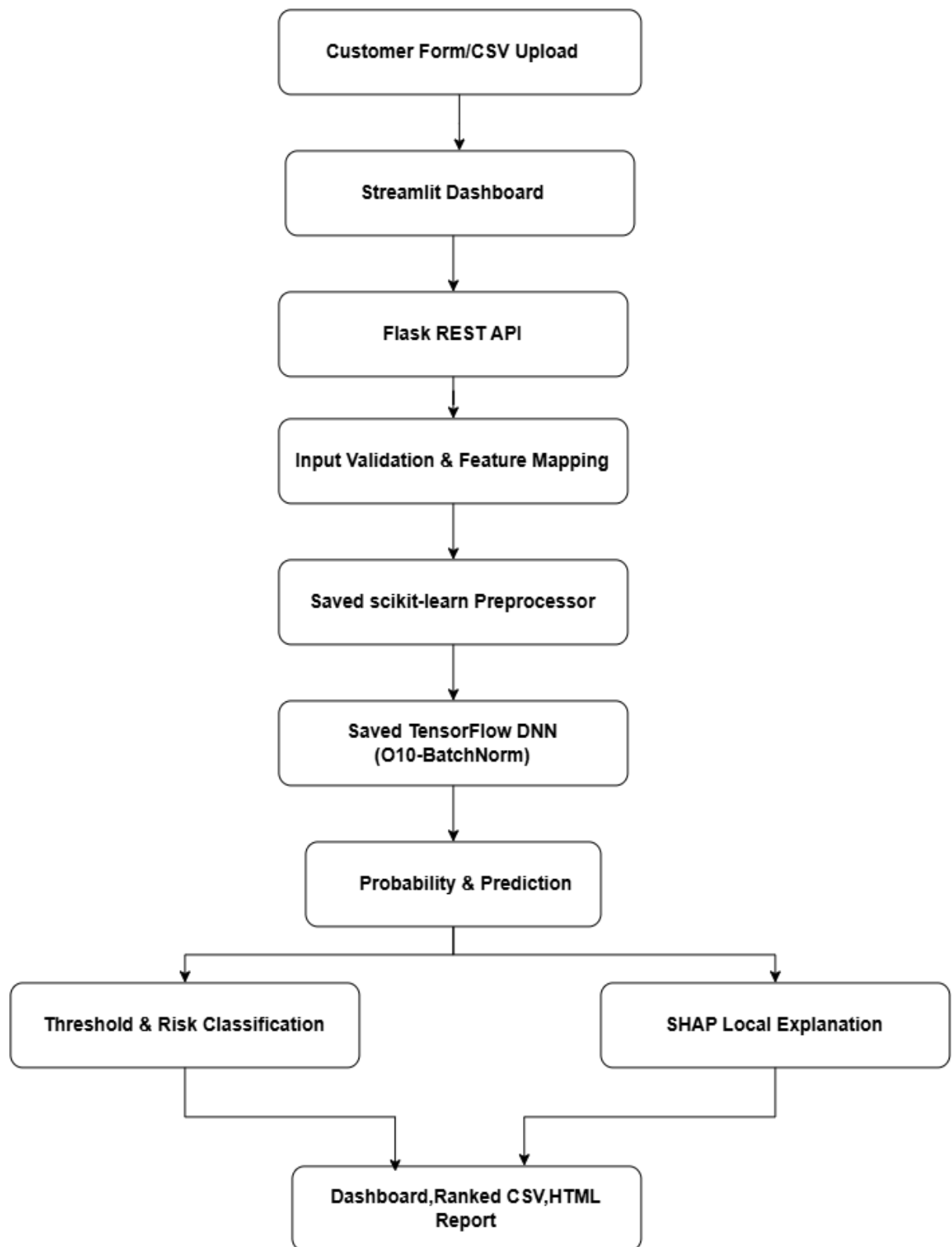


Figure 4.1: Application architecture of the churn intelligence system

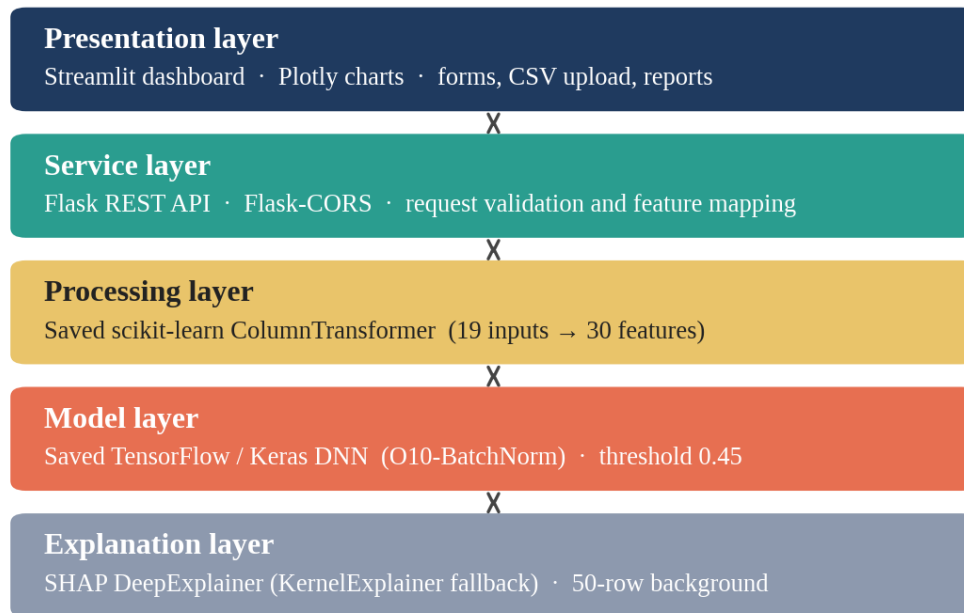


Figure 4.2: Layered view of the system

Table 4.1 lists what each layer contains and what it is responsible for.

Layer	Components	Responsibility
Presentation	Streamlit dashboard, Plotly charts	Collects input (form or CSV), displays predictions, explanations, insights and reports
Service	Flask REST API, Flask-CORS	Exposes the endpoints, validates requests and maps fields to the expected schema
Processing	Saved scikit-learn ColumnTransformer (preprocessor_optimized.joblib)	Imputes, scales and encodes the 19 raw fields into 30 model inputs
Model	Saved TensorFlow/Keras DNN (churn_dnn_optimized.keras)	Produces a churn probability between 0 and 1
Decision	Threshold and risk-band logic	Turns the probability into a predicted label and a Low, Medium or High risk level

Layer	Components	Responsibility
Explanation	SHAP DeepExplainer with KernelExplainer fallback, 50-row background matrix	Produces local (per customer) and global (portfolio) feature contributions

Table 4.1: Layers of the Proposed System

Key design decisions

- **Fit once, reuse always.** The preprocessor and the model are saved as artifacts and loaded once per API process. Training-time and inference-time transformations therefore always match, and requests do not pay the cost of reloading large files.
- **A thin front end.** Streamlit only displays results. The interface can be changed without touching the model logic, and the model can be updated without rewriting the interface.
- **A tunable operating point.** The threshold and risk bands live in the backend configuration and environment variables, so a retention team can change them without retraining.
- **Explanations on demand.** A batch run scores every row, but a SHAP explanation is computed only for the row an analyst selects. This keeps large uploads fast.

4.2 High Level Design

At a high level, the system does five things: it collects customer data, prepares it, predicts churn, explains the prediction, and reports the result. We grouped this work into the modules shown in Table 4.2, each mapped to the functional requirement it satisfies. Each module is designed to perform a specific task independently while communicating seamlessly with other modules to ensure accurate and efficient processing. This modular design simplifies system maintenance, supports future enhancements, and improves the overall scalability and reliability of the customer churn prediction system.

Table 4.2: Functional Modules and Requirement Mapping

Module	What it does	Requirement
Data input and validation	Accepts a form entry or a CSV file; checks required fields, data types, allowed category values and file structure	FR-01
Preprocessing	Fixes blank TotalCharges, imputes missing values, scales numeric fields and encodes categorical fields	FR-02
Churn prediction	Runs the DNN and applies the 0.45 threshold to label a customer Likely or Not Likely to churn	FR-03
Risk classification	Assigns Low, Medium or High risk from the predicted probability	FR-03
Explainability	Computes local SHAP contributions per customer and global mean absolute SHAP values	FR-04
SHAP Assistant	Answers plain-language questions about the latest prediction and its explanation	FR-05
Evaluation and insights	Shows stored performance metrics, the baseline comparison and global feature importance	FR-06
Reporting	Produces ranked CSV files and downloadable HTML reports for human review	FR-03, FR-05

The Flask API exposes one endpoint for each part of this workflow, listed in Table 4.3. Keeping them separate lets the dashboard ask for exactly what it needs on each page. This endpoint-based design improves communication between the frontend and backend by allowing data to be exchanged efficiently and securely. It also makes the system easier to maintain, test, and extend with additional features or services in the future.

Table 4.3: API Endpoints and Their Purpose

Endpoint	Purpose	Used by
Health and feature metadata	Confirms the API and model are available and returns the expected 19-field input schema	Dashboard start-up, monitoring
Single-customer prediction	Returns probability, predicted label, threshold and risk level	Predict page
Prediction with explanation	Returns the prediction and its SHAP explanation in one call, saving a round trip	Predict page
Batch CSV prediction	Scores every customer in an uploaded file and returns them ranked by risk	Batch scoring
Batch-row explanation	Computes a SHAP explanation for one selected row of a scored batch	Batch scoring
Model insights (/api/model/insights)	Returns stored performance metrics and global feature importance	Insights page

Usage scenarios

Scenario A: scoring one customer.

1. The analyst fills in the customer form on the Predict page.
2. The dashboard calls the prediction-with-explanation endpoint.
3. The API validates the record, applies the saved preprocessor and runs the DNN.
4. The response carries the probability, predicted label, threshold, risk level and the strongest SHAP drivers.
5. The analyst reads the result, asks the SHAP Assistant a follow-up question if needed, and downloads a report.

Scenario B: ranking a customer portfolio.

1. The analyst uploads a CSV file exported from the billing system.
2. The API checks the structure of the file and validates every row.
3. Each customer is scored, then the list is sorted by descending churn probability.

4. The analyst downloads the ranked CSV or an HTML report for the retention team.
5. For any customer of interest, the analyst requests a SHAP explanation for that single row.

4.3 Detailed Design

4.3.1 Data preprocessing pipeline

Raw customer records cannot go straight into a neural network, so a fixed sequence of preprocessing steps is applied (Table 4.4). The whole sequence is one scikit-learn ColumnTransformer, fitted once on the training data and saved as preprocessor optimized.

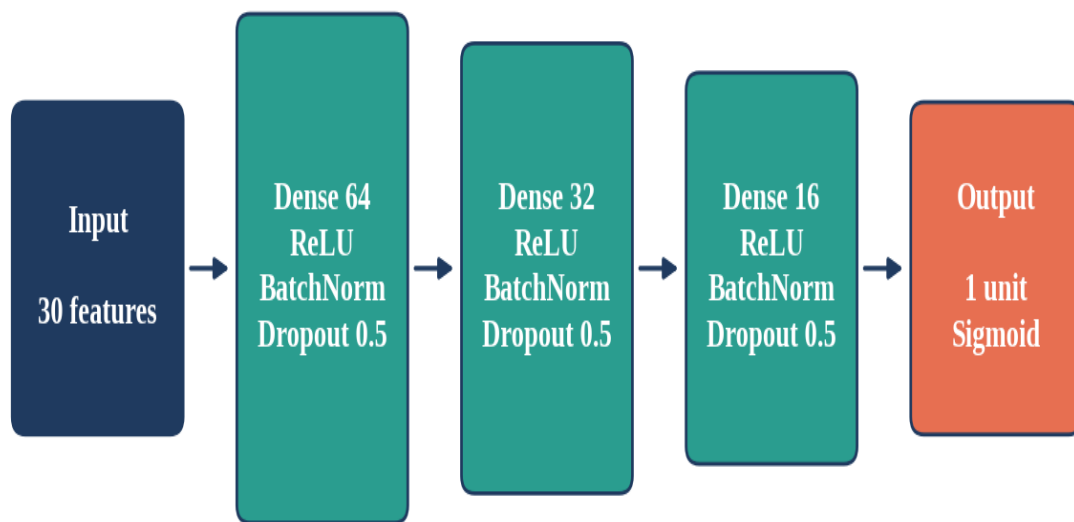
Table 4.4: Preprocessing Steps and Rationale

Step	Operation	Why it is done
1	Convert whitespace-only TotalCharges values to missing numbers	A few customers with zero tenure have no billing history yet, and the blank string cannot be parsed as a number
2	Impute missing numeric values with the median	Charges are right-skewed, and the median is less affected by extreme values than the mean
3	Impute missing categorical values with the most frequent category	Keeps every record usable without inventing a new category
4	Standardise numeric variables with StandardScaler	Puts tenure and charges on a comparable scale, which helps gradient-based training converge
5	One-hot encode categorical variables, dropping the first category	Avoids perfectly correlated dummy columns and keeps SHAP feature names readable
6	Output 30 transformed features	Fixed input size for the neural network

Saving and reusing this fitted transformer is an important choice. If the API refitted or rebuilt the preprocessing on its own, small differences could silently change the meaning of the features the model receives, and predictions would drift without any visible error.

4.3.2 Deep neural network

The production model is a network named O10-BatchNorm, the tenth in a documented series of architecture experiments. It takes the 30 transformed features and passes them through three hidden blocks. Each block is a dense layer with ReLU activation, followed by batch normalization and a dropout rate of 0.5. A single sigmoid unit at the end outputs the churn probability. Figure 4.3 shows the structure and Table 4.5 lists the layers with their parameter counts.



Adam optimizer ($lr = 0.0003$) | Binary cross-entropy | Batch size 32 | Early stopping (patience 10)

Figure 4.3: Structure of the O10-BatchNorm deep neural network

Table 4.5: Layer Summary of the DNN

Layer	Output shape	Parameters
Input (30 features)	30	0
Dense, 64 units, ReLU	64	1,984
Batch normalization	64	256
Dropout (0.5)	64	0
Dense, 32 units, ReLU	32	2,080
Batch normalization	32	128
Dropout (0.5)	32	0
Dense, 16 units, ReLU	16	528
Batch normalization	16	64
Dropout (0.5)	16	0
Output dense, 1 unit, sigmoid	1	17
Total		5,057 (4,833 trainable, 224 non-trainable)

Parameter counts are calculated from the layer sizes listed above.

Two design choices deserve a short explanation. Batch normalization keeps the distribution of activations steady from one layer to the next [31], which makes training less sensitive to the learning rate. Dropout switches off a random half of the neurons during each training step [32], so the network cannot lean too heavily on a few co-adapted units. With roughly 5,600 training rows (under an 80/20 split) and a few thousand parameters, overfitting is a real risk, and both techniques help guard against it.

Table 4.6: Training Configuration

Setting	Value
Optimizer	Adam [30], learning rate 0.0003
Loss function	Binary cross-entropy
Batch size	32
Early stopping	Patience of 10 epochs on the monitored validation metric
Hidden activation / output activation	ReLU / sigmoid
Regularisation	Batch normalization and dropout (0.5) after every hidden layer
Decision threshold	0.45

4.3.3 Decision threshold and risk policy

The sigmoid output is a score between 0 and 1. A separate policy converts it into a decision. A customer is predicted to churn when the score is 0.45 or higher, rather than at the usual 0.50. We chose the lower value on purpose: in a retention setting, missing a customer who really was about to leave costs more than contacting one who would have stayed. Lowering the threshold trades some precision for extra recall.

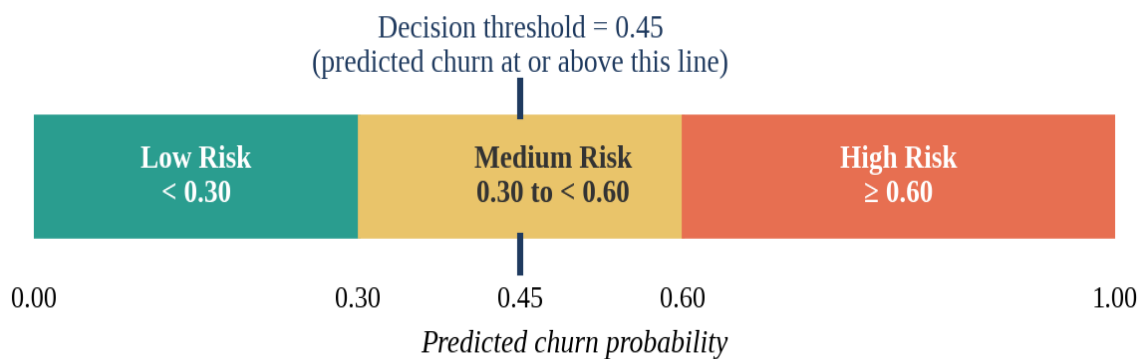
**Figure 4.4: Decision threshold and risk bands**

Table 4.7: Risk Bands and Suggested Handling

Risk level	Predicted probability	Suggested handling
Low	Below 0.30	Monitor; no active outreach
Medium	0.30 up to, but not including, 0.60	Lower-effort campaign, for example an automated email offer
High	0.60 or above	Immediate review and personal outreach

The threshold and the band limits are read from the backend configuration and environment variables. A retention team can therefore change the operating point when its capacity or priorities change, without retraining the model.

4.3.4 Explainability module

Explanations are produced with SHAP [24]. The application loads a saved background matrix of 50 transformed rows that acts as the reference sample. It first tries SHAP's DeepExplainer, which is designed for differentiable models and uses the network's gradients to compute contributions efficiently. If the installed TensorFlow and SHAP versions turn out to be incompatible, it falls back to the model-agnostic KernelExplainer. This fallback exists because the two libraries are released on independent schedules and do not always work together.

The module produces two kinds of output.

- **Local explanation (one customer).** For each feature it returns the contribution and its direction, the size of the impact, the baseline value it is measured against, and the strongest churn-driving and retention-supporting factors for that customer. Positive values push the prediction toward churn and negative values push it toward staying.
- **Global importance (whole portfolio).** The mean absolute SHAP value of each feature across the observations examined. A feature counts as important if it moves predictions a lot, in either direction, across many customers. The backend computes this over the first 20 rows of the cached background data, which keeps it fast enough to run on demand.

4.3.5 Input validation and reporting

Every request is validated before it reaches the model. The API checks that all required features are present, that values have the right data type, that categorical values are among

the allowed options, and that an uploaded CSV has the correct structure. Anything that fails is rejected with a clear message, so the system never quietly produces a prediction from malformed input. The customerID column is kept only to identify records and is never passed to the model.

For batch runs, results are sorted by descending churn probability and returned with the predicted label and risk level. They can be downloaded as a ranked CSV file or as an HTML report that a retention team can read without further processing.

4.4 Datasets Descriptions

The project uses the IBM Telco Customer Churn dataset [25], stored in the file WA_Fn-UseC_-Telco-Customer-Churn.csv. It has 7,043 customer records and 21 columns: 19 input attributes, the customerID identifier and the Churn label. The attributes describe four things about each customer: who they are, which services they subscribe to, how they are billed, and how long they have been with the provider. Table 4.8 lists them.

Table 4.8: Attributes of the Telco Customer Churn Dataset

Group	Attribute	Type	Values / description
Demographic	gender	Categorical	Male, Female
	SeniorCitizen	Numeric (0/1)	Whether the customer is a senior citizen
	Partner, Dependents	Categorical	Yes, No
Account	tenure	Numeric	Months the customer has stayed with the company
	Contract	Categorical	Month-to-month, One year, Two year
	PaperlessBilling	Categorical	Yes, No
	PaymentMethod	Categorical	Electronic check, Mailed check, Bank transfer (automatic), Credit card (automatic)

Group	Attribute	Type	Values / description
	MonthlyCharges, TotalCharges	Numeric	Current monthly bill and total billed to date
Phone and internet	PhoneService	Categorical	Yes, No
	MultipleLines	Categorical	Yes, No, No phone service
	InternetService	Categorical	DSL, Fiber optic, No
Add-on services	OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies	Categorical	Yes, No, No internet service
Target	Churn	Categorical	Yes (churned) or No (retained)
Identifier	customerID	Text	Used to identify records only; excluded from model inputs

Of the 7,043 customers, 1,869 churned and 5,174 stayed, so about 26.54% of the customer base was lost (Figure 4.5). This imbalance matters. A model that always predicted “stays” would already be right about three times in four, which is why accuracy alone is a poor yardstick here and why we report precision, recall, F1-score and ROC-AUC as well.

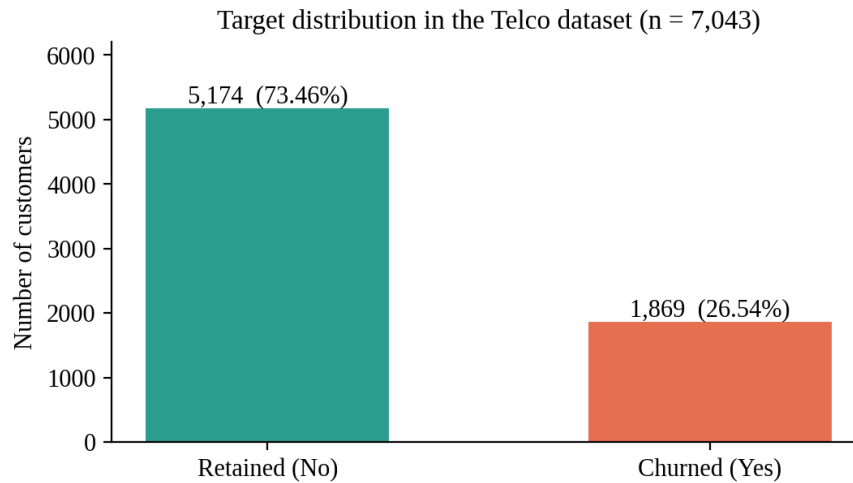


Figure 4.5: Class distribution of the Churn label

After preprocessing, the four numeric inputs (SeniorCitizen, tenure, MonthlyCharges, TotalCharges) and the one-hot encoded categorical inputs together form the 30 features the network receives. The dataset comes from a single provider's residential customers at one point in time. It is a good benchmark for showing a complete workflow, but it should not be read as a picture of churn in every telecom market.

4.5 Tools and Technologies used for Implementation

The whole system is written in Python and runs on an ordinary CPU-based machine without GPU acceleration. Python versions 3.9 to 3.13 are supported, subject to the installed TensorFlow, SHAP, and scikit-learn versions, because DeepExplainer compatibility can differ between releases. Table 4.9 lists the main software tools and technologies used for system development and deployment. The software environment was selected to ensure compatibility, ease of development, and efficient execution on standard desktop or laptop computers. This configuration enables the system to perform customer churn prediction, explainability analysis, and report generation without requiring specialized hardware or high-performance computing resources.

Table 4.9: Tools and Technologies

Category	Tool	Used for
Language	Python	All backend, modelling and dashboard code [34]
Deep learning	TensorFlow / Keras	Defining, training and running the neural network [26]
Machine learning	scikit-learn	Preprocessing pipeline and baseline models [27]
Baseline model	XGBoost	Gradient-boosting comparison model [35]
Data handling	pandas, NumPy, joblib	Data manipulation, array computation and saving fitted artifacts [33]
Explainability	SHAP	DeepExplainer and KernelExplainer feature attributions [24]
Backend API	Flask, Flask-CORS	REST endpoints, validation and cross-origin access [28]
Frontend	Streamlit	Interactive dashboard [29]
Visualisation	Plotly	Churn-distribution and feature-importance charts

CHAPTER 5

Results and Discussion

This chapter reports how the system performed. It describes the experimental set-up, presents the results for each functional and non-functional requirement, compares the work with the studies from the literature survey, and closes with the societal impact and the Sustainable Development Goals the project supports.

5.1 Experimentation Environment Set up

All experiments ran on a CPU-based machine; no GPU was needed. The software stack is Python with TensorFlow/Keras, scikit-learn, SHAP, pandas, NumPy and joblib. Exact library versions matter here, because SHAP's DeepExplainer support depends on the installed TensorFlow version. Table 5.1 summarises the set-up.

Table 5.1: Experimental Set-up

Item	Setting
Dataset	IBM Telco Customer Churn, 7,043 records, 19 input attributes
Evaluation split	Stratified 80/20 split, random_state = 42 (about 5,634 training and 1,409 test customers)
Preprocessing	Saved ColumnTransformer (preprocessor_optimized.joblib), 19 inputs to 30 features
Model	O10-BatchNorm DNN (churn_dnn_optimized.keras), 64-32-16 hidden units, dropout 0.5
Training	Adam, learning rate 0.0003, binary cross-entropy, batch size 32, early-stopping patience 10
Decision threshold	0.45
Explainer	SHAP DeepExplainer, 50-row saved background matrix
Baselines	Logistic Regression, Decision Tree, Random Forest, XGBoost
Hardware	CPU-only machine, no GPU acceleration

Two sets of performance figures appear in this chapter, and it helps to be clear about where each comes from. The first are the metrics **stored by the application itself** (its `/api/model/insights` endpoint) for the deployed model. The second come from an **independent re-evaluation** that we ran using only the artifacts shipped with the application. We did this because the original training notebook and its exact split were not available in the project files. For the re-evaluation, the saved preprocessor transformed the held-out rows exactly as it does at inference time, the saved Keras model produced probabilities, and the production threshold of 0.45 turned them into predictions.

5.2 Functional Requirements Results achieved

This section takes the functional requirements from Chapter 3 one at a time and shows what the system delivers for each.

5.2.1 Data input, validation and preprocessing (FR-01, FR-02)

The system accepts the 19 customer fields through a form or a CSV upload and validates them before anything reaches the model. Validation covers required fields, data types, allowed category values and file structure. Accepted records are cleaned and transformed by the saved preprocessor into the 30 features the network expects. Because the same fitted pipeline is reused for every request, the transformation at inference time is identical to the one used in training.

5.2.2 Churn prediction and risk classification (FR-03)

For a single customer the system returns a churn probability, a predicted label (Likely to Churn or Not Likely to Churn), the threshold used and a risk level of Low, Medium or High. For a CSV upload it returns every customer with the same fields, sorted from highest to lowest churn probability, together with a downloadable ranked CSV file and an HTML report. A retention team can therefore start from the top of the list and work down as far as its capacity allows.

5.2.3 Model performance (FR-06)

Table 5.2 gives the metrics stored by the application for the deployed model at a threshold of 0.45. Table 5.3 gives the results of the independent re-evaluation on the 20% test split.

Table 5.2: Stored Project Metrics (threshold 0.45)

Metric	Score
Accuracy	78.21%
Precision	57.81%
Recall	66.31%
F1 Score	61.77%
ROC-AUC	83.97%

Table 5.3: Independent Evaluation (stratified 80/20 split, random_state = 42, threshold 0.45)

Metric	Score
Accuracy	79.70%
Precision	62.29%
Recall	59.63%
F1 Score	60.93%
ROC-AUC	84.19%

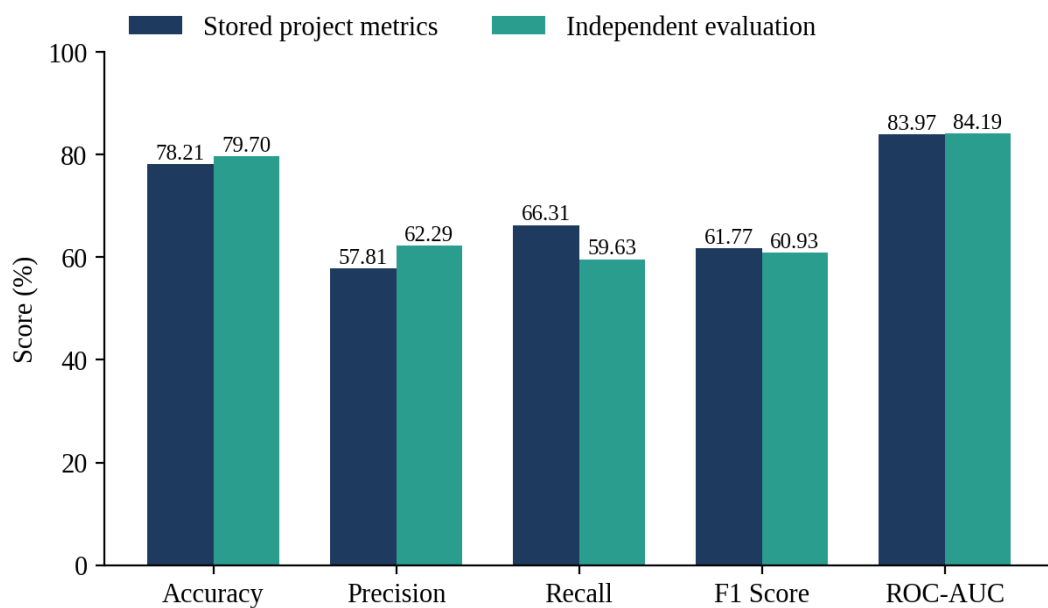


Figure 5.1: Stored metrics compared with the independent evaluation

The two sets of figures do not match exactly. The independent run shows higher accuracy (79.70% against 78.21%) and precision (62.29% against 57.81%) but lower recall (59.63% against 66.31%), while ROC-AUC is almost unchanged (84.19% against 83.97%). Both runs use the same saved model and preprocessor, so neither result is wrong. The most likely explanation is that the split, the preprocessing fit or the way the threshold was chosen in the original notebook differs from our fixed-seed protocol.

The pattern fits what we would expect. ROC-AUC does not depend on the threshold, and it barely moved. Recall shifted by almost seven points because, with only 26.54% churners, a test set holds roughly 374 positive cases, and recall is sensitive to exactly which customers land in it. The practical lesson is that any reported metric should be quoted together with its split, seed and threshold. In a future release the exact split indices, seeds, package versions and a metrics file should be saved alongside the model so the discrepancy cannot recur.

The confusion matrix for the independent evaluation is shown in Figure 5.2 and Table 5.4. The counts were reconstructed from the reported accuracy, precision and recall and the size of the test split (1,409 customers, about 374 of whom churned); they reproduce all three reported figures exactly.

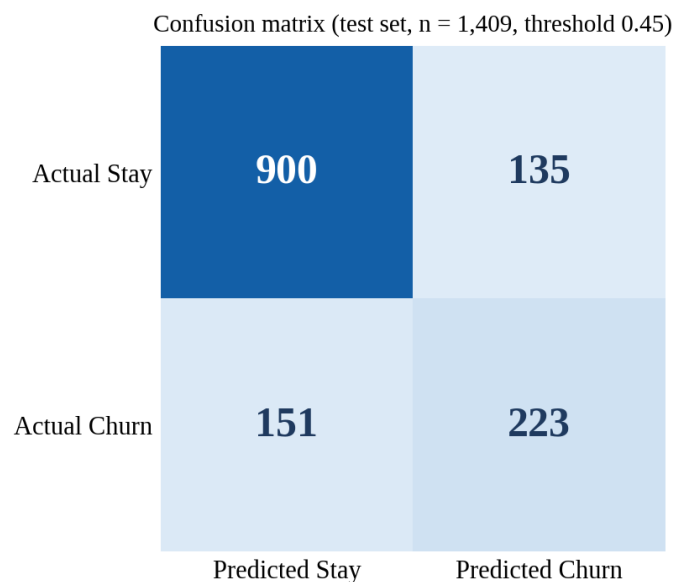


Figure 5.2: Confusion matrix of the independent evaluation

Table 5.4: Confusion Matrix Counts and Derived Measures (independent evaluation)

Measure	Value	Meaning
True positives	223	Churners correctly flagged
False positives	135	Loyal customers flagged unnecessarily
False negatives	151	Churners the model missed
True negatives	900	Loyal customers correctly left alone
Specificity	86.96%	Share of loyal customers correctly identified
Lift over random	about 2.3×	Precision of 62.29% against a base churn rate of 26.54%

In practical terms, the model flags 358 customers, about a quarter of the test set, and 223 of them really were about to leave. Picking the same number of customers at random would catch roughly 95. The 135 unnecessary contacts are the price of catching 223 churners, and for an outreach that costs little and rarely harms the relationship, that is a reasonable trade.

Table 5.5 and Figure 5.3 compare the deployed DNN with four baseline models using the stored comparison report. These figures come from the project’s historical evaluation, so they should be read together with the note on split provenance above.

Table 5.5: Baseline Model Comparison (stored figures)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	80.55%	65.82%	55.61%	60.29%	84.20%
Decision Tree	79.42%	62.96%	54.55%	58.45%	82.84%
Random Forest	78.92%	63.05%	49.73%	55.61%	82.26%
XGBoost	80.20%	65.68%	53.21%	58.79%	84.45%
Optimized DNN	78.21%	57.81%	66.31%	61.77%	83.97%

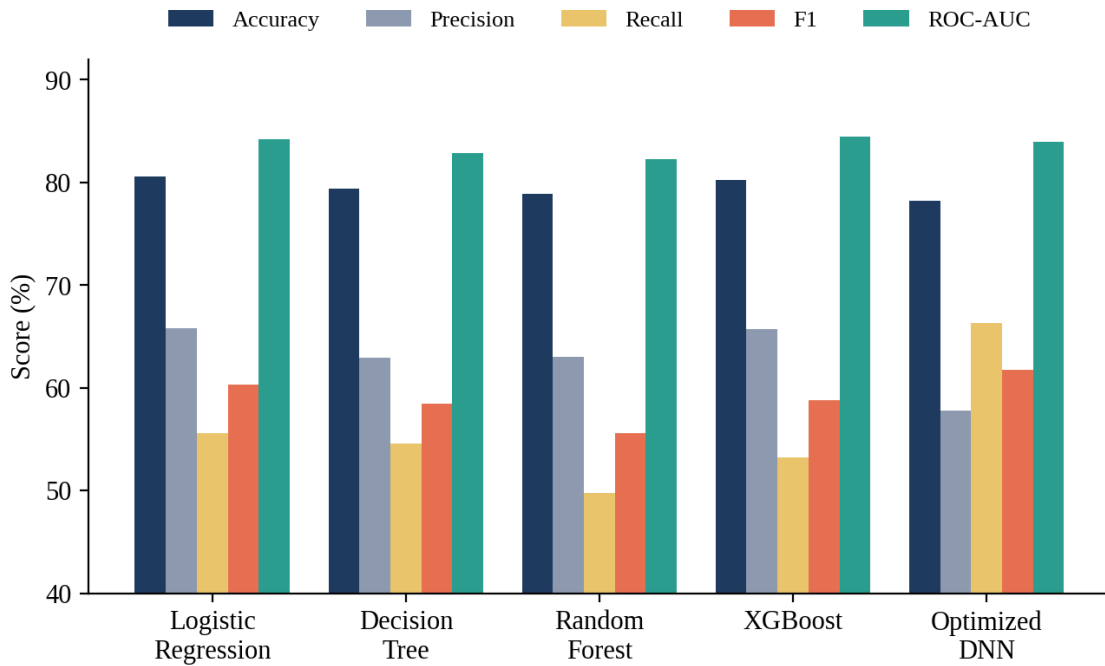


Figure 5.3: Baseline comparison across five metrics

The optimized DNN has the highest recall and F1-score of the five models, but Logistic Regression and XGBoost are ahead on accuracy and slightly ahead on ROC-AUC. The difference is not marginal. The DNN's recall of 66.31% is more than ten points above Logistic Regression's 55.61%, while its accuracy is only about two points lower. That is exactly the behaviour the 0.45 threshold was chosen to produce. For retention work, where overlooking a likely churner is usually the costlier mistake, the extra recall can be worth more than a small gain in accuracy.

A ROC-AUC near 0.84 means the model separates churners from non-churners well above chance, but well short of the near-perfect separation needed to treat its output as a final decision. The system is therefore best used to rank and triage customers, with a person deciding what to do next.

5.2.4 Explainable AI analysis (FR-04)

Global importance was recomputed with DeepExplainer over the first 20 rows of the saved 50-row background matrix, and the transformed features were ranked by mean absolute SHAP value. Table 5.6 and Figure 5.4 show the top ten.

Table 5.6: Top Ten Global SHAP Features

Rank	Feature	Mean absolute SHAP
1	tenure	0.066268
2	MonthlyCharges	0.064695
3	InternetService_Fiber optic	0.037602
4	Contract_Two year	0.033984
5	PaymentMethod_Electronic check	0.029402
6	TotalCharges	0.028331
7	PaperlessBilling_Yes	0.021484
8	TechSupport_Yes	0.019074
9	OnlineSecurity_Yes	0.019009
10	MultipleLines_Yes	0.017686

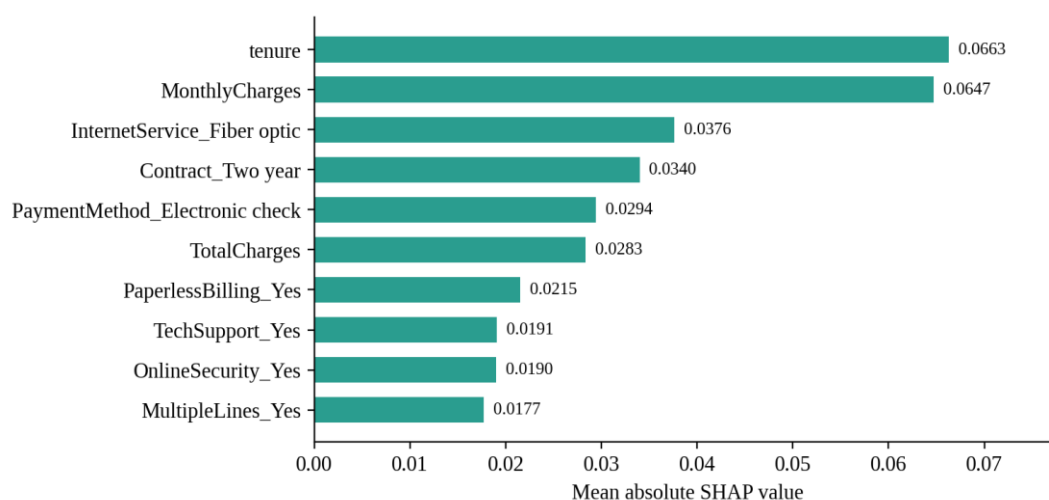


Figure 5.4: Top ten features by mean absolute SHAP value

Tenure and monthly charges stand clearly apart from the rest, which suggests that where a customer is in their lifecycle and how much they pay matter more to the model than any other single attribute. Contract type, internet service, payment method, paperless billing and the support and security subscriptions add secondary signals.

These rankings need to be read with care. SHAP describes what the **model** relies on; it does not prove that changing a feature would change a customer's behaviour. A short tenure, for

instance, may simply stand in for early dissatisfaction that the data does not record. The rankings are also specific to this model, this dataset and this background sample. They are useful for deciding where to look and which retention experiments to try, not as statements about what causes churn.

5.2.5 SHAP Assistant and reports (FR-05)

The dashboard includes an assistant that answers plain-language questions about the most recent prediction and its SHAP explanation. It surfaces the strongest churn-driving and retention-supporting factors for that customer, so an analyst can tailor the conversation: a pricing review for a customer whose risk is tied to high monthly charges, or a service-quality follow-up where the risk is tied to fiber-optic service. The advice is limited to what the SHAP drivers support. Richer, offer-level recommendations are left for future work.

5.2.6 Summary of functional requirements

Table 5.7: Functional Requirements Status

ID	Requirement	Outcome	Status
FR-01	Customer data input	Form and CSV input with validation of fields, types, categories and file structure	Achieved
FR-02	Data preprocessing	Saved pipeline turns 19 inputs into 30 features consistently	Achieved
FR-03	Churn prediction	Probability, label at 0.45 and Low/Medium/High risk; ranked batch output	Achieved
FR-04	Explainable AI	Local and global SHAP explanations with feature importance	Achieved
FR-05	Chatbot assistance	SHAP Assistant explains results in plain language; recommendations limited to SHAP drivers	Achieved (basic)
FR-06	Performance evaluation	Accuracy, precision, recall, F1, ROC-AUC and confusion matrix reported	Achieved

5.3 Non-Functional Requirements Results achieved

Non-functional requirements are harder to express as a single number, so Table 5.8 records how each one is met and what evidence supports it.

Table 5.8: Non-Functional Requirements Status

ID	Requirement	How it is met	Evidence / result
NFR-01	Performance	Model, preprocessor and explainer are initialised once per process; batch scoring runs in one pass; SHAP for a batch is computed per selected row	Design-level; response-time benchmarks were not measured in this study
NFR-02	Reliability and reproducibility	Fitted preprocessor and model are saved and reused; independent re-evaluation with a fixed split	Reproduced the model's ROC-AUC (84.19% against 83.97%); metric discrepancy documented
NFR-03	Usability	Forms, plain-language assistant, ranked tables, downloadable CSV and HTML reports	Achieved
NFR-04	Data integrity	Validation of required fields, types, categories and CSV structure; customerID excluded from inputs	Achieved
NFR-05	Maintainability	Separate presentation, API, preprocessing, model and explanation layers; threshold and risk bands set through configuration	Achieved
NFR-06	Portability	Runs on CPU only with Python 3.9 to 3.13	Achieved; library versions must be compatible

ID	Requirement	How it is met	Evidence / result
NFR-07	Transparency	Local and global SHAP output with every prediction; explanations described as model behaviour, not cause	Achieved

Beyond these individual checks, three qualities stand out. **Reproducible inference**: reusing the saved artifacts removes a common and often silent source of error in deployed machine-learning systems. **Decision-oriented design**: the threshold, the risk bands and the recall emphasis all reflect the retention use case rather than a default setting. **Operational coverage**: single scoring, batch ranking, reports and health checks are part of the system itself, not extras added to a notebook. The one gap is speed: we did not run formal latency or load tests, and we list that as future work.

5.4 Comparison of Results with Existing works

Table 5.9 places this work next to the studies from the literature survey that report comparable information. The figures come from different datasets, preprocessing choices, and evaluation set-ups, so the table gives context rather than a head-to-head ranking. This comparison highlights how the proposed system performs relative to existing customer churn prediction approaches while acknowledging differences in experimental conditions. It also demonstrates that the proposed model achieves competitive performance by combining accurate prediction with explainable AI and an AI chatbot for improved decision support.

Table 5.9: Comparison with Existing Work

Study	Data and method	Reported result	Explainability
Khan et al. [21]	ANN on telecom customer attributes	About 79% accuracy	Attrition factors identified
Nalatissifa and Pardede [4]	Tuned DNN on Telco churn data	Competitive with, not automatically better than, classical models	Not reported
Rasool et al. [22]	Stacking ensemble with SMOTE on a different telecom dataset	94.75% accuracy, 73.20% recall, 80.23% F1	SHAP
El Attar and El-Hajj [23]	Seven ML models with SMOTE on the same Telco dataset	Accuracy and F1 of 0.84; XGBoost AUC 0.932	SHAP and customer segmentation
Saha et al. [16]	ChurnNet deep neural network	Improved over traditional ML models	Lacking
This work	Optimized DNN on the Telco dataset, no resampling, deployed as an API and dashboard	79.70% accuracy, 59.63% recall, AUC 84.19% (independent); 78.21% accuracy, 66.31% recall (stored)	Local and global SHAP

Three observations follow. First, the accuracy of this work sits in the same range as the ANN study of Khan et al. [21], even though the datasets and networks differ. Second, the stronger figures reported in [22] and [23] were obtained with oversampling and, in [22], on a different dataset. Our figures come from an untouched, imbalanced test split with no resampling, so they should not be set directly against theirs. Third, the pattern in the baseline comparison, in which the DNN gains recall but not accuracy, agrees with the conclusion of [4] that a neural network is not automatically better than classical models.

The feature-level results also line up with earlier work. El Attar and El-Hajj [23] used the same dataset and found contract type, tenure and technical support to be the most influential

features. Our ranking places tenure first, two-year contracts fourth and technical support eighth. The overlap suggests that these patterns come from the data itself rather than from one network or one training run.

Where this project differs most from the existing studies is not in headline accuracy but in scope. Most of the reviewed papers concentrate on a model and a metrics table. This project connects a validated API, an interactive dashboard, batch prioritisation, downloadable reports and SHAP explanations into one tool that a retention team can actually use.

5.5 Project GUI Snapshots (Major Functionalities)

The Streamlit dashboard is organised around the tasks a retention analyst performs. The screens below show its major functions. Replace each frame with the corresponding screenshot from the running application.

The screenshot displays the 'Telecom Churn Intelligence' dashboard. At the top, there's a header with the title and a subtitle: 'Score individual customers, review the factors behind each prediction, perform batch scoring, and prioritize retention work with confidence using explainable AI.' Below the header is a navigation bar with tabs: 'Overview', 'Predict Churn' (active), 'Batch Customers', and 'SHAP Assistant'. The main section is titled 'SINGLE CUSTOMER RISK EVALUATOR'. It contains a form with various input fields and dropdown menus for customer data. The fields are organized in a grid-like structure. At the bottom of the form is a 'Calculate Risk Score' button.

Customer Name / Reference	Internet Service Type	Gender
Manaswi	Fiber optic	Female

Contract Type	Payment Method	Partner
Month-to-month	Electronic check	No

Tenure (Months)	Paperless Billing	Dependents
12	Yes	No

Monthly Charges (\$)	Senior Citizen Status	Multiple Lines
75.50	No (0)	No

Total Charges (\$)	Phone Service	Online Security
906.00	Yes	No

> Additional Services

Calculate Risk Score

Figure 5.5: Predict page: customer form and single-customer result with churn probability and risk level

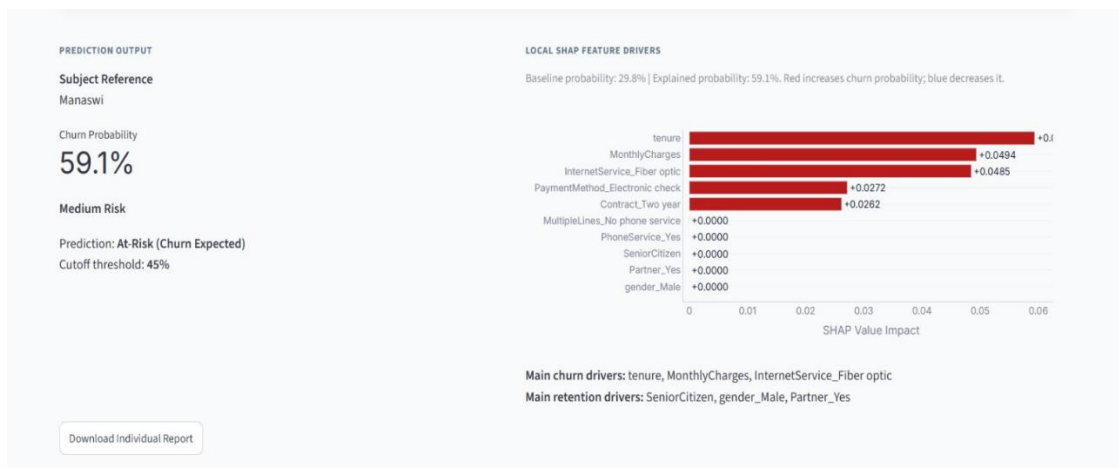


Figure 5.6: Local SHAP explanation for one customer, showing the strongest churn-driving and retention-supporting factors

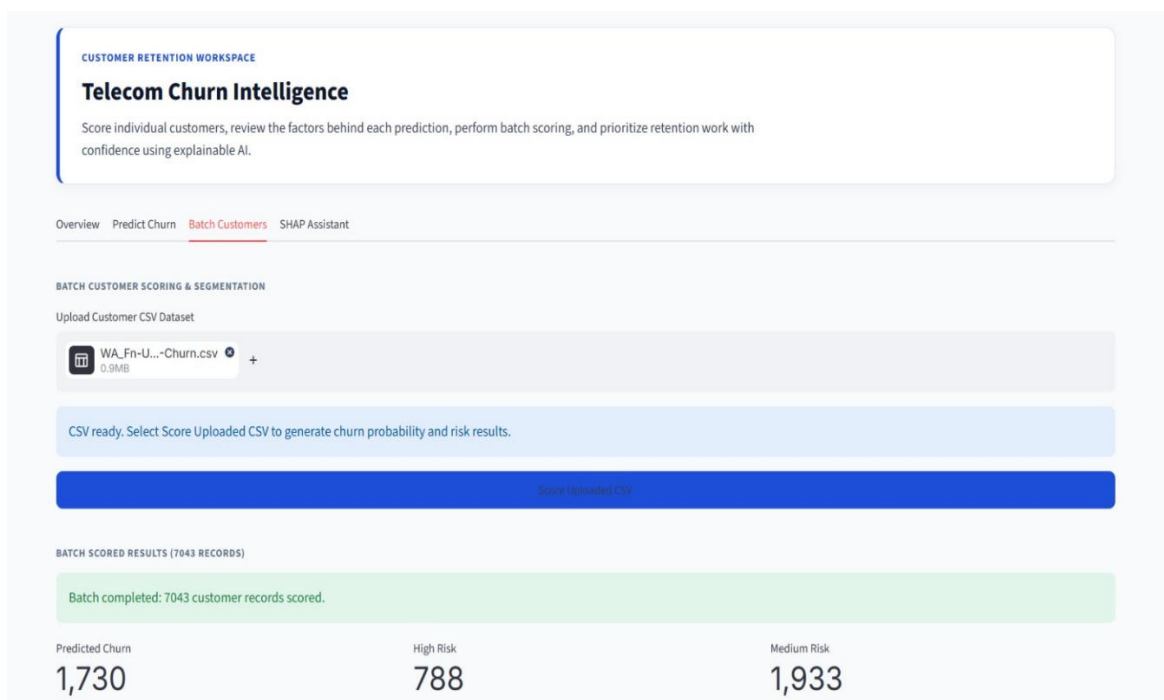


Figure 5.7: Batch scoring: uploaded CSV ranked by churn probability with risk levels and download options

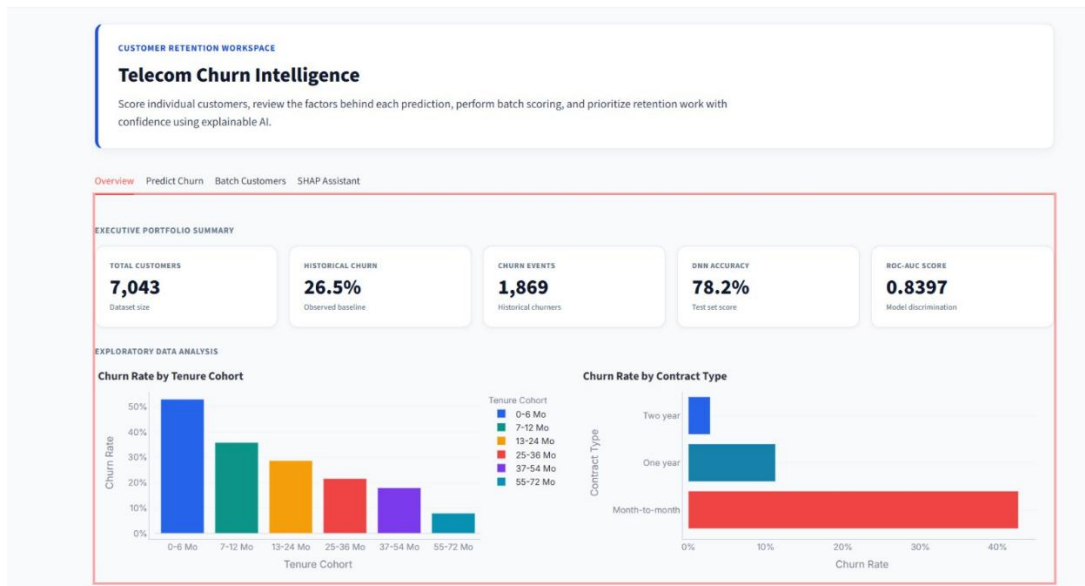


Figure 5.8: Insights page: stored model metrics, baseline comparison and global SHAP feature importance

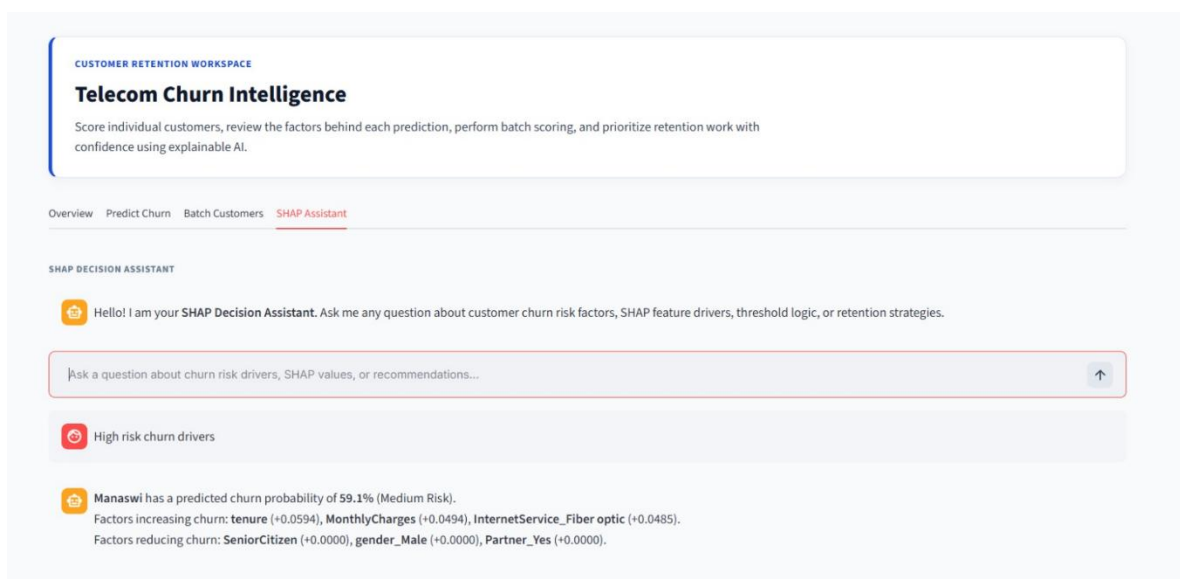


Figure 5.9: SHAP Assistant answering a plain-language question about the latest prediction

5.6 Societal Impact of the Project

Churn prediction can look like a purely commercial tool, but it touches real people on both sides of the relationship, and a few of these effects are worth stating plainly.

- **Better service for customers.** When a provider knows early that someone is at risk, it can fix a billing problem, review a plan or follow up on a service issue before the

customer gives up. Explanations make the outreach relevant: a customer with a high monthly bill hears about pricing, not a generic offer.

- **Sustainable businesses and jobs.** Keeping existing customers protects revenue that pays for network upgrades and staff. The system runs on ordinary CPU hardware, so smaller operators can use it without a large infrastructure budget.
- **Trust through transparency.** Because every prediction comes with reasons, staff can justify why a customer was contacted and question the model when its reasoning looks wrong. This supports accountable, human-supervised use of AI.
- **Responsibilities that come with the tool.** The model uses personal and billing data, so privacy and secure handling are essential. Inputs such as gender and senior-citizen status call for a fairness review before any real deployment. The output should be used to help customers, for example with better offers or support, and never to justify worse treatment without human review.

5.7 SDG Mapped

The project contributes most directly to the following United Nations Sustainable Development Goals.

Table 5.10: Sustainable Development Goals Mapped to the Project



Conclusion and Future Scope

Conclusion

This project built an end-to-end system for predicting telecom customer churn and explaining its predictions. A saved scikit-learn pipeline turns 19 customer attributes into 30 model inputs, a regularised deep neural network estimates the churn probability, a 0.45 threshold and three risk bands turn it into action, and SHAP shows why. A Flask API and a Streamlit dashboard make all of this usable for single customers and for whole portfolios, with ranked CSV files and HTML reports for the retention team.

In the project's own stored evaluation the model reached 78.21% accuracy, 57.81% precision, 66.31% recall, 61.77% F1 and 83.97% ROC-AUC. An independent re-run of the same saved artifacts on a fixed stratified 80/20 split gave 79.70% accuracy, 62.29% precision, 59.63% recall, 60.93% F1 and 84.19% ROC-AUC. We report both sets because the difference is itself a finding: it shows why every metric needs a documented split and protocol. Global SHAP analysis ranked tenure, monthly charges, fiber-optic internet, two-year contracts and electronic-check payment as the strongest influences, in line with earlier work on the same dataset.

The system is a decision-support tool, not an autonomous decision-maker. It is meant to help people decide who to call first and what to talk about, and it should stay under calibration, monitoring, fairness review and human oversight as it moves into wider use.

Limitations

- The dataset covers one provider's residential customers at a single point in time, so the results may not carry over to other countries, products or enterprise customers.
- Only about a quarter of the customers churned, so accuracy alone is misleading; results depend on the chosen threshold and on the split used to measure them.
- The 0.45 threshold reflects assumed retention priorities and should be recalibrated against an organisation's real outreach capacity, costs and customer lifetime value.
- The stored and independent metrics do not match exactly because the original notebook and split were not preserved with the model.
- SHAP values describe what the model has learned, not what causes churn, and they can be affected by correlated variables and by the background sample.

- Response time and load behaviour were not measured.

Future Scope

- **Frozen evaluation artifacts.** Save split indices, seeds, package versions, training history and a metrics file with each model version.
- **Probability calibration.** Test Platt scaling or isotonic regression before treating the sigmoid output as a true probability of churn.
- **Cost-sensitive thresholds.** Choose the threshold and risk bands from customer lifetime value, outreach cost and agent capacity instead of fixed values.
- **Temporal validation.** Test the model on later customer periods to see how it behaves as behaviour drifts.
- **Monitoring after deployment.** Track missing data, category drift, score distributions, and recall, precision and calibration over time.
- **Fairness analysis.** Compare error rates and calibration across sensitive groups, where legally and ethically appropriate.
- **Intervention experiments.** Link predictions to controlled retention campaigns and measure the retention actually gained.
- **Wider model comparison.** Benchmark calibrated gradient boosting and simpler interpretable models on the same frozen split, and check how stable the SHAP ranking is across background samples and retrained models.
- **Speed testing and richer assistance.** Measure latency under load, and extend the assistant to suggest concrete retention offers.

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